Guide to the Model

A general equilibrium model of multinational production, market power, and trade policy

Project brief

August 17, 2026

Part I gives the structure of the model. **Part II** builds the full general equilibrium model. **Part III** solves the model step by step. **Part IV** explains how the model is matched to customs data. **Part V** is what the model is for: who owns the profits, and what it says about tariffs. **Part VI** lists the calibration and then asks, fact by fact, whether the model actually reproduces the six stylized facts. **Part VII** is limits, a glossary, and where to find things.
Everything here is reproduced by a single file, `mne_model.jl`; `julia mne_model.jl quick` runs it in about two minutes.

Contents

Part I
The model in three equations

1 The whole thing

Three main equations:

$$\text{(1) What firms charge:} \quad \frac{1}{\mu_g} = 1 - \frac{1-S_g}{\sigma} - \frac{S_g}{\eta} \tag{1}$$

$$\text{(2) What they earn:} \quad \Pi_g = \frac{E}{\sigma} S_g \left[1 + \left(\tfrac{\sigma}{\eta} - 1\right) S_g\right] \tag{2}$$

$$\text{(3) Shares add up:} \quad \sum_g S_g = 1 \tag{3}$$

A *market* is one destination country buying one product — say *Brazil buying auto parts.* Inside it, a handful of companies compete. The more of the market a company controls, the more it can charge. That sentence is equations (1) and (2).

Everything else in this document is: where these come from, what determines the things they take as given (E and the costs behind S_g), and why the whole system has exactly one solution.

2 Equation 1: what a firm charges

$$\boxed{\frac{1}{\mu_g} = 1 - \frac{1-S_g}{\sigma} - \frac{S_g}{\eta}}$$

Symbol	Name	Meaning
g	a *group*	One company, counting all its subsidiaries together.
μ_g	the *markup*	How many times its cost the group sells for.
S_g	*market share*	The slice of the market's spending it captures.
σ	within-product substitution	How easily buyers switch *brands.*
η	across-product substitution	How easily buyers switch *products.*

The markup. If a unit costs \$10 to make and deliver and sells for \$18, then $\mu_g = 1.8$. It is simply price divided by cost, so price = $\mu_g \times$ cost.

The two elasticities. These say how picky buyers are. σ asks: *if one brand of auto part gets a prince increase of 1%, how many buyers switch to a rival brand?* η asks: *if auto parts as a whole gets a price increase of 1%, how much do buyers spend on something else entirely?* The second is much smaller, since you cannot really substitute away from auto parts.

2.1 Reading it

In words: How much a firm can charge above cost depends on how big it is.

The right-hand side splits the market in two. On the fraction $1 - S_g$ the group does **not** own it fights rivals, and the relevant pickiness is σ. On the fraction S_g it **does** own, pushing for more sales just steals from itself, so the only question is whether buyers leave the product category altogether — that is η. The formula is a weighted average of the two.

2.2 The one idea that makes the model work

When a company sells one more unit it pushes the price down a little. A *small* company hardly cares: that price cut applies to its few units. A *big* company cares a lot: it applies to *all* of its many units. So big companies hold back on purpose, and that is why they charge more.

Economists call this *cannibalisation* — expanding eats your own sales.

2.3 Numbers you can check

With $\sigma = 5$ and $\eta = 1$, equation (1) becomes

$$\mu_g = \frac{\sigma}{(\sigma-1)(1-S_g)} = \frac{5}{4\,(1-S_g)}. \tag{4}$$

Share S_g	Markup	Mark-on	Reading
0%	1.25	25%	a tiny firm; the textbook answer
10%	1.39	39%	
30%	1.79	79%	a big player
50%	2.50	150%	half the market
90%	12.50	1150%	nearly a monopolist

Check one: at $S_g = 0.5$, $\mu = 5/(4 \times 0.5) = 2.5$.

2.4 The two extremes, and what each one tells us

A very small firm. Set $S_g = 0$: $\mu = \sigma/(\sigma-1)$. This is the standard textbook markup, and it does not involve η at all — a firm too small to matter cannot move the whole product category. **This is our benchmark:** it is what a conventional trade model predicts for every firm.

A monopolist. Set $S_g = 1$: $\mu = \eta/(\eta-1)$. If $\eta = 1$ this is *infinite*, which is exactly what $\eta = 1$ means (buyers spend a fixed amount whatever the price). Section 7 explains why we still use $\eta = 1$ much of the time and how we guard against it.

3 Equation 2: what a firm earns

$$\boxed{\Pi_g = \underbrace{\frac{E}{\sigma}S_g}_{\text{textbook part}} + \underbrace{\frac{E}{\sigma}\left(\frac{\sigma}{\eta}-1\right)S_g^2}_{\text{bonus for being big}}}$$

E is *market size*, the total money spent on this product in this destination. In Part II it stops being a given number and becomes something the model determines.

In words: Profit is what any ordinary model would predict, plus extra profit that exists *only* because the firm is big.

That squared term is the seed of the whole project. Added across firms, $\sum_g S_g^2$ is the **Herfindahl index**, the standard measure of market concentration. So the "extra" profit in a market is proportional to how concentrated it is. Ask *who owns* it and you get an *ownership-weighted* Herfindahl. That object is the paper.

With $E = \$100\text{m}$, $\sigma = 5$, $\eta = 1$:

Share	Textbook part	Bonus part	Total	Bonus as % of textbook
5%	\$1.0m	\$0.2m	\$1.2m	20%
30%	\$6.0m	\$7.2m	\$13.2m	120%
60%	\$12.0m	\$28.8m	\$40.8m	240%

In words: For a firm with 30% of a market, a conventional model reports \$6m of profit. The truth is \$13.2m. It misses more than half.

4 Equation 3: shares add up

$$\boxed{\sum_g S_g \;=\; 1}$$

It is what ties all the firms together, since each firm's share depends on what everyone else does. Part III is about the fact that there is **exactly one** way to satisfy it.

Part II

The full model, ingredient by ingredient

5 What "general equilibrium" means, and why we need it

A *partial* equilibrium model takes some things as given from outside: market size, wages, costs. A *general* equilibrium model determines everything inside the model, and insists that every market clears at once.

Concretely, a general equilibrium model must have all six of these. The rest of Part II is one section per ingredient.

	Ingredient	What it answers
1	Endowments	What does the world start with?
2	Demand	Who wants what, and with how much money?
3	Supply	How are goods made, and with whose labour?
4	Conduct	How are prices set?
5	Ownership	Who receives the profits?
6	Clearing	What must balance for everything to be consistent?

Why we do it this way. Earlier versions of this project were partial equilibrium: E and the costs c_i were fixed numbers typed in from outside. That was fine for understanding markups, but it made the central policy question unanswerable. With costs fixed, a tariff cannot change any foreign supplier's cost, so there is no *terms-of-trade* effect at all — and the model then says the best policy is always an import *subsidy*, whatever the ownership structure. The general equilibrium closure is what lets wages, and therefore foreign costs, respond.

6 Ingredient 1: endowments

There are N countries. Country n has L_n units of labour, and labour is the only primitive factor of production. Its price is the wage w_n, which the model determines.

Why we do it this way. One factor keeps the model transparent and is standard in this literature. Capital, land, and intermediates could be added, but each brings its own market that must clear, and none of them is needed for the ownership question. What matters here is that *something* must be scarce, so that a tariff which shifts demand between countries moves relative prices. Labour does that job.

7 Ingredient 2: demand

There are K sectors (think: HS6 product categories). Country n spends its entire income X_n, in two stages.

Stage 1: across sectors. Spending is split with elasticity η and weights β_k:

$$E_{nk} \;=\; \underbrace{\beta_k \left(\frac{P_{nk}}{P_n}\right)^{1-\eta}}_{\equiv\ \varepsilon_{nk}} X_n, \qquad P_n \;=\; \left(\textstyle\sum_k \beta_k P_{nk}^{1-\eta}\right)^{\frac{1}{1-\eta}}. \tag{5}$$

The shares ε_{nk} add to one automatically, so all income is spent. When $\eta = 1$ this collapses to $E_{nk} = \beta_k X_n$: fixed budget shares.

Stage 2: within a sector. Buyers like variety. Given quantities $q_1, q_2, \ldots$ of the competing varieties, satisfaction is

$$Q \;=\; \Big(\sum_j q_j^{\rho}\Big)^{1/\rho}, \qquad \rho \;\equiv\; \tfrac{\sigma-1}{\sigma} \in (0,1). \tag{6}$$

Why we do it this way. Two reasons for the two-tier CES structure.

Two tiers because switching *brand* must be easier than switching *product*. Two brands of the same auto part are near-substitutes; an auto part and a banana are not. One tier cannot express that. This is why we require $\sigma > \eta$ throughout — an economic statement, not a technical one. If it failed, the model would predict that bigger firms charge *lower* markups, which is backwards, and (as Section 15 shows) the solution would stop being unique.

CES mainly for honesty. Under plain CES with many small firms, **ownership does not matter at all**. So this functional form hands us our null hypothesis for free. Had we chosen a form that builds the ownership effect in, the result would be worthless. Instead we recover "ownership is irrelevant" exactly in the limit of many small firms, and everything we report is a measured departure from it.

Where this comes from. The two-tier CES with large firms is the market structure of **Atkeson and Burstein (2008)**. The nesting matters: an outer tier across sectors with a low elasticity η, an inner tier across varieties with a high elasticity σ. That gap is what lets a firm be small in the economy but *large in its own market* — which is the whole point, since a firm that is large in its own market has a markup that depends on its share.

The single-tier CES that Ramondo–Rodríguez-Clare and Tintelnot use is the special case in which every firm is negligible. We nest their case rather than contradict it: as $S_g \to 0$ our markup becomes their constant $\sigma/(\sigma-1)$. That is deliberate, and it is what makes the null hypothesis honest.

Why we do it this way. *On $\eta = 1$ specifically.* It is convenient: budget shares are fixed and the algebra is cleanest. But it is a knife-edge exactly where our story lives — at $\eta = 1$ a firm with the whole market has an infinite markup and absorbs a tariff completely. Both are artefacts. So the code accepts any η, and every headline number is re-checked at $\eta > 1$, where a monopolist's markup is the sensible $\eta/(\eta-1)$. As Section 9.4 shows, the ownership result is unchanged: only a single constant moves.

7.1 What demand implies about prices and shares

From (6), if buyers spend E in total, the price variety i commands is

$$p_i \;=\; \frac{E\, q_i^{-1/\sigma}}{A}, \qquad A \;\equiv\; \sum_j q_j^{\rho}, \tag{7}$$

and its share of spending is

$$s_i \;=\; \frac{p_i q_i}{E} \;=\; \frac{q_i^{\rho}}{A}. \tag{8}$$

In words: The more of variety i on the market, the cheaper it is; and the more of *everything else*, the cheaper it is too. That is the A in the denominator — think of A as "how crowded this market is".

Shares add to one automatically, since $\sum_i q_i^{\rho} = A$. So equation (3) is built into demand. A group's share is the sum over the varieties it owns, $S_g = \sum_{i \in g} s_i$.

8 Ingredient 3: supply — multinational production

This is the part that makes it a model *of multinationals*, and it is where the Ramondo–Tintelnot framework enters.

An *affiliate* a is a plant. It has three addresses and one number:

$g(a)$	the **parent**: the global ultimate owner
$h(a)$	the parent's **headquarter country**
$\ell(a)$	the country the plant **produces in**
φ_a	the plant's productivity

Its cost of delivering one unit into destination n, in sector k, is

$$\boxed{\underbrace{a_{a,n}}_{\text{before tariff}} = \frac{1}{\varphi_a} \underbrace{\left(w_{h(a)}^{\alpha_k}\, w_{\ell(a)}^{1-\alpha_k}\right)^{1-\nu_k}}_{\text{(i) HQ + production labour}} \underbrace{\left(P^{\text{IO}}_{\ell(a),k}\right)^{\nu_k}}_{\text{(ii) intermediates}} \underbrace{\gamma_{h(a),\,\ell(a)}}_{\text{(iii) MP friction}} \underbrace{d_{\ell(a),\,n}}_{\text{(iv) shipping}}}$$

$$\boxed{c_{a,n} = a_{a,n}\big(1 + t_{\ell(a),n,k}\big).} \quad (9)$$

Take the four pieces one at a time.

(i) $w_h^{\alpha_k} w_\ell^{1-\alpha_k}$: whose labour actually makes the good. A multinational uses **two** countries' labour. A fraction α_k of cost is *headquarter services* — design, brand, management, R&D — paid at the *parent's* wage. The remaining $1 - \alpha_k$ is production, paid at the *host's* wage.

Where this comes from. This is the **headquarter input** of Head and Mayer (2019), the third borrowed piece.

What it is. Studying cars, they observe that where a model is *assembled* and where the brand *comes from* both affect its cost and its appeal. Some of what a product costs is attributable to the parent — design, engineering, brand, management — and that part is produced at home and does not relocate when assembly does. α_k is the size of that part.

Why the model needs it. It is what makes a multinational genuinely *multi*-national. Without it an affiliate would be an ordinary local firm that happens to have a foreign owner, and ownership would have no effect on cost at all. With it, an affiliate is a genuine hybrid: part parent, part host.

Three things it buys us.

- It is the natural home for **complexity** (Stylized Fact 2): α_k rises with product complexity, since complex goods need more headquarter input.
- It gives tariffs a surprising incidence: a tariff on a US-owned Mexican affiliate falls partly on *American* headquarter workers.
- It makes the affiliate's cost depend on *two* wages, which is what connects the ownership question to the labour market.

An honest complication, and it is instructive. As written, α_k alone does **not** deliver Fact 2 — it pushes the wrong way. A foreign affiliate pays its *parent's* wage on the α_k share, and parents sit in high-wage countries, so raising α_k makes foreign ownership more *expensive.* Part VI shows this explicitly and shows what the fix is: the headquarter input has to be a **capability local firms cannot buy at any price**, not merely costly labour. Fact 2 is what tells us that.

(ii) $P^{\mathrm{IO}}_{\ell k}$: the inputs a firm buys from other firms. A fraction ν_k of cost is not labour at all — it is intermediate goods, bought *in the country where the plant sits*, at the prices prevailing there:

$$P^{\mathrm{IO}}_{\ell k} \;=\; \prod_{k'} \big(P_{\ell k'}\big)^{\omega_{k'k}}, \qquad \sum_{k'} \omega_{k'k} = 1. \tag{10}$$

$\omega_{k'k}$ is the share of sector k's input bill spent on sector k' goods — an input–output table.

Where this comes from. This is the **roundabout production** structure of Caliendo and Parro (2015).

What it is. They point out that most of what a factory buys is not labour but *other firms' output*: steel, chemicals, components, services. In national accounts, intermediates are roughly half of gross output. So sectors are chained together, and a cost shock anywhere propagates through the chain.

Why the model needs it. Without it, a multinational arriving in a country can only ever *take* business from local firms. With it, it also *sells* to them — and cheaply, since multinationals are the productive firms. This is the only channel in the model through which multinationals can help the firms around them, so it is the channel Stylized Fact 5 needs. Part VI reports how far it gets, which is honestly not far enough on its own.

How it is disciplined. ν_k is the intermediate share of gross output and ω comes from an input–output table; both are directly observable. We use $\nu = 0.55$ and a diagonal-heavy ω.

What it costs us. It creates a genuine circularity — cost depends on prices, prices depend on cost — which is the one new solution step in the model. Section 16 proves that loop has exactly one solution.

In words: When a multinational sets up in Colombia it sells there too, which lowers $P_{\ell k'}$, which lowers the input bill of *every* Colombian firm. Local exporters get cheaper inputs.

(iii) $\gamma_{h\ell}$: the cost of operating abroad. A parent from h running a plant in ℓ loses some efficiency: $\gamma_{hh} = 1$ at home and $\gamma_{h\ell} > 1$ abroad.

Where this comes from. This is the **multinational production friction** of Ramondo and Rodríguez-Clare (2013), and it is the single most important thing we borrow.

What it is in their model. They ask what a country gains from letting foreign technologies be used at home, not just foreign goods be imported. A firm's know-how originates in one country and can be *used* in another, but not costlessly: moving a technology abroad degrades it. $\gamma_{h\ell}$ is the size of that degradation — how much efficiency an h-technology loses when operated in ℓ. It plays the same role for multinational production that an iceberg trade cost plays for trade, which is why the two enter our cost function (8) side by side and multiplicatively.

Why the model needs it. Without γ, ownership and location would be the same decision. Every firm would simply produce wherever wages are lowest, and "a US firm producing in Mexico" would be indistinguishable from "a Mexican firm". γ is precisely what makes *who owns the plant* a different question from *where the plant is* — and since this project is about ownership, without it there would be no project.

What it does here. A high $\gamma_{h\ell}$ discourages h-parents from operating in ℓ, so it governs how much multinational activity there is and where. That makes it the natural lever for Stylized Facts 1 and 3 — how large the multinational share is, and which countries parents come from.

How it is disciplined. In their work it is inferred from data on multinational production relative to trade. We set it to a single number, $\gamma = 1.18$ for all foreign pairs, which is a simplification we make openly: it is enough to generate multinational activity of a realistic size, but it cannot by itself explain *which* countries become parents. That is why Fact 3 is an input here rather than a result, and why making γ bilateral and estimating it is the route to turning Facts 1 and 3 into model output.

(iv) $d_{\ell n}$: shipping. An iceberg cost from the *production* country to the destination. "Iceberg" means that to deliver one unit you must ship $d_{\ell n} \geq 1$ units and the rest melts away — a standard trick that turns transport costs into a simple cost multiplier.

Where this comes from. The fact that ℓ and n are **separate indices** is the **export platform** structure of Tintelnot (2017), and it is the second thing we borrow.

What it is in his model. Earlier multinational models mostly assumed a firm opens a plant in a country *to sell in that country.* Tintelnot's point is that this is not what firms do: a German carmaker's plant in Mexico exists to serve North and South America. So a plant chosen in ℓ becomes a *platform* from which many destinations n are served, and the firm's location decision depends on the whole map of destinations it wants to reach.

Why the model needs it. Two reasons, and both are central here.

First, this project is about affiliates in Latin America that *export.* If we forced $\ell = n$, an affiliate could only sell where it sits, there would be no affiliate exports at all, and the question "where do multinational exports go?" — which decides the sign of our main policy result — could not even be asked.

Second, it is what generates Stylized Fact 6 without being told to. Because distance runs from the *production* country rather than from headquarters, a multinational serves each destination from whichever of its plants is cheapest. Its sales are therefore much less sensitive to distance from home than an ordinary exporter's. That is exactly the pattern Fact 6 reports, and Part VI shows the model reproduces both its signs.

How it is disciplined. We set $d_{\ell n} = \text{distance}^{1/(\sigma-1)}$. That exponent is not free: it makes the elasticity of *trade* with respect to distance exactly -1, the central estimate of the gravity literature. See Section 27.

What we do not take from him. Tintelnot lets firms *choose* their plant locations, paying fixed costs. We take the set of locations as observed. That caveat is discussed under Ingredient 4.

The tariff. $t_{\ell nk}$ is charged on the *customs value*, so it multiplies the pre-tariff cost. This is why $a_{a,n}$ and $c_{a,n}$ are tracked separately: the difference is government revenue, not a payment to any worker.

This is where general equilibrium bites. In equation (8) the marginal cost contains *wages*. Wages are determined by labour market clearing. So when a country imposes a tariff, demand for foreign labour falls, foreign wages fall, and foreign costs fall. That is the *terms-of-trade* effect, and it simply does not exist when costs are fixed numbers.

9 Ingredient 4: conduct — who competes with whom

Inside each market, **groups** choose quantities, taking rivals' quantities as given.

Why we do it this way. *Why the parent rather than the plant.* This is the most important modelling decision in the project. A parent that owns three exporters of the same product does not let them undercut each other. Counting them as three independent firms understates both concentration and market power. The data agree: grouping affiliates by parent raises measured concentration (Stylised Fact 4).

Why we do it this way. *Why quantities (Cournot) rather than prices (Bertrand).* It is the convention in this literature, so our numbers are comparable to others'; it gives a clean closed form; and quantity competition fits goods made in fixed plants with capacity limits. Bertrand would give a similar-looking but different markup rule, and belongs in a robustness appendix rather than being hidden.

Where this comes from. The share-dependent markup itself is **Atkeson and Burstein (2008)**. They study a nested CES market in which firms are large enough to affect their own price index, which is what makes the markup depend on market share rather than being a constant. Equation (1) is their markup rule, applied to *groups* of affiliates rather than to single firms.

This is where we depart from Ramondo–Tintelnot, and the departure is the contribution.
In Ramondo–Rodríguez-Clare there is perfect competition. In Tintelnot there is monopolistic competition with a **constant** markup. In both, no firm is large enough to matter on its own.
That is fatal for our question. With constant markups the S_g^2 term of equation (2) is **identically zero**: ownership washes out of welfare almost entirely and market concentration has no role at all. Their models cannot express the sentence this project is about.
So we keep their *entire* multinational production structure — who owns what, where it produces, who it sells to, the MP friction, the headquarter input, the export platforms, the general equilibrium closure — and change exactly one assumption: firms compete as groups, and they are big. That single change is what makes concentration, ownership and market power interact.

9.1 One honest caveat: we do not choose locations

Tintelnot (2017) lets firms *choose* which countries to produce in, paying fixed costs. We take that set as observed in the data.

Why we do it this way. Combining endogenous location choice with oligopoly is genuinely hard. With tariffs in the payoff, the mathematical property that makes the location problem tractable can fail, and no existence proof is available. So we take locations as given and treat re-sourcing as a robustness exercise with an elasticity borrowed from that literature.

The cost is specific and worth naming: it means Fact 6 is *reproduced* rather than *explained*. The model shows that multinationals face flatter distance gradients given where their plants are; it does not explain why those plants are there. This is a real limitation and it is stated as one.

9.2 Deriving equation (1)

A group picks quantities for its own varieties to maximise

$$\Pi_g \;=\; \underbrace{E\,S_g}_{\text{revenue}} \;-\; \underbrace{\sum_{i\in g} c_i q_i}_{\text{cost}}\,. \tag{11}$$

There is exactly one calculus step in the whole model. Differentiate S_g using (8):

$$\frac{\partial S_g}{\partial q_i} = \underbrace{\frac{\rho\, q_i^{\rho-1}}{A}}_{\text{my share rises}} - \underbrace{\frac{\left(\sum_{j\in g} q_j^{\rho}\right)\rho\, q_i^{\rho-1}}{A^2}}_{\text{but the market grew}} = \frac{\rho\, q_i^{\rho-1}}{A}\big(1-S_g\big).$$

In words: Shipping more raises my share, but it also enlarges the whole market, which dilutes my share. The $(1-S_g)$ is exactly that dilution — and it shrinks the bigger I already am. That is cannibalisation, in algebra.

Now the useful coincidence: $\rho - 1 = -1/\sigma$, so $q_i^{\rho-1} = q_i^{-1/\sigma}$, and comparing with (7) gives $E\, q_i^{\rho-1}/A = p_i$. Hence marginal revenue is $\rho\, p_i(1 - S_g)$. Setting it equal to marginal cost c_i:

$$\boxed{p_i \;=\; c_i\, \mu_g, \qquad \mu_g \;=\; \frac{\sigma}{(\sigma-1)(1-S_g)}} \tag{12}$$

which is equation (4). Derived, not assumed.

> Look hard at (12): the markup contains S_g, the **group's total share**, not the individual variety's share. So **every affiliate of the same parent charges the same markup**, however big or small each one is. Their *prices* differ because their costs differ, but price-over-cost is common across the group.
> This is a sharp, testable prediction, and it is why grouping affiliates by parent raises measured concentration *and* true market power at the same time.

9.3 Deriving equation (2)

Rearranging (12), each variety's cost is revenue divided by the markup, so $\sum_{i \in g} c_i q_i = E S_g/\mu_g$. Substituting into (11) gives $\Pi_g = E S_g(1 - 1/\mu_g)$, and since

$$1 - \frac{(\sigma-1)(1-S_g)}{\sigma} = \frac{1 + (\sigma-1)S_g}{\sigma},$$

we get $\Pi_g = (E/\sigma)\, S_g\, [1 + (\sigma-1)S_g]$: equation (2) with $\eta = 1$.

Sanity checks. $S_g \to 0$ gives $E S_g/\sigma$, the textbook answer. $S_g \to 1$ gives $\Pi_g \to E$: the monopolist captures all the spending, which is right when $\eta = 1$.

9.4 The general case

Redoing the same steps with $\eta \neq 1$ gives exactly equations (1) and (2).

> Compared with $\eta = 1$, the *only* thing that changes is that the constant $(\sigma - 1)$ becomes $(\sigma/\eta - 1)$. The structure — textbook part plus a bonus proportional to S_g^2 — is identical. The central result therefore does **not** rest on the convenient special case.

10 Ingredient 5: ownership

Let θ_{gn} be the fraction of parent g owned by residents of country n, with $\sum_n \theta_{gn} = 1$ for every parent: all profits go to somebody.

Where this comes from. That ownership of foreign firms should change a country's optimal tariff is an old idea, going back to **Brecher and Bhagwati (1981)** and developed by **Blanchard** and co-authors: if you own the firms you are taxing, taxing them takes from your own pocket, so you tax them less.

What is new here is not the idea but the object. Every paper in that line uses a *country-level* ownership share — one number for how much of the foreign economy a country owns. We have global-ultimate-parent identity matched to individual customs transactions, so θ_{gn} is attached to each *firm*. Part V shows why that distinction is not cosmetic: two situations with the identical country-level share and the identical market concentration can differ by 31% in what actually matters.

Why we do it this way. Ownership is a *global* object attached to the parent, not to the plant. That is the whole point. A US-owned affiliate in Colombia sends its profits to the United States regardless of which market it sells into. This is the single piece that country-level models cannot represent, because they only have an average ownership share, not the identity of which firms are owned.

11 Ingredient 6: what must balance

Three conditions, and one consequence.

(a) Goods markets clear. In every (n, k) cell, $\sum_g S_{g,nk} = 1$. Already built into demand.

(b) Labour markets clear. In every country m, labour demanded equals L_m. Labour is demanded from two directions, which is the multinational part:

$$w_m L_m = \sum_{a,n} \Big[\underbrace{\alpha_k \, a_{a,n} q_{a,n} \, \mathbf{1}\{h(a) = m\}}_{\text{HQ services, parent's country}} + \underbrace{(1 - \alpha_k) \, a_{a,n} q_{a,n} \, \mathbf{1}\{\ell(a) = m\}}_{\text{production, host country}} \Big]. \tag{13}$$

(c) Every country's budget balances.

$$X_n = \underbrace{w_n L_n}_{\text{wages}} + \underbrace{\sum_g \theta_{gn} \Pi_g}_{\text{profits it owns, worldwide}} + \underbrace{T_n}_{\text{tariff revenue it collects}} . \tag{14}$$

Income equals spending. Tariff revenue is rebated to consumers as a lump sum — the government is not a separate agent.

The consequence: Walras' law and the numeraire. Add equation (14) over all countries and it turns out one clearing condition is implied by the others. Formally, the value of excess labour demand is identically zero at *every* wage vector:

$$\sum_m w_m \big(\text{labour demanded}_m - L_m\big) = 0. \tag{15}$$

Two practical consequences, and one reassurance.
Consequences. Only $N - 1$ of the labour market conditions are independent, so we may drop one; and only relative wages are determined, so we set $w_1 = 1$. Neither loses anything.
Reassurance. Equation (15) is the single best test that the accounting is complete, because it must hold **out of equilibrium too**. If we had lost track of a single euro — forgotten that iceberg shipping costs are paid to workers, or double-counted tariff revenue — it would fail. We check it at randomly chosen, deliberately non-equilibrium wages. It holds to 2.6×10^{-16}.

12 Provenance

Each piece was sourced where it was introduced. Here they are on one page, so you can see at once how much is borrowed and how little is new — which is the point.

Piece	Source	What it does here
Two-tier CES, large firms	Atkeson–Burstein (2008)	lets a firm be small in the economy but large in its market
Parent in h, plant in ℓ, sales to n	Ramondo–Rodríguez-Clare (2013)	the basic multinational structure
MP friction $\gamma_{h\ell}$	Ramondo–Rodríguez-Clare (2013)	makes *who owns* different from *where*
Plant serves many destinations	Tintelnot (2017)	export platforms; generates Fact 6
HQ input share α_k	Head–Mayer (2019)	makes an affiliate part parent, part host
Intermediates ν_k, $\omega_{k'k}$	Caliendo–Parro (2015)	lets multinationals *supply* local firms
Ownership lowers the tariff	Brecher–Bhagwati (1981), Blanchard	the question we are asking
GE closure	standard	wages, clearing, budgets
Group-level Cournot	**ours**	**makes concentration and ownership interact**
Firm-level θ_{gn}	**ours**	**what the ORBIS/D&B match buys**

Nine of the eleven rows are borrowed, and deliberately so: the multinational production block is not where we are trying to be original. The contribution is the last two rows, and they only mean something *because* the rest is standard.

Part III

Solving the model, and why there is exactly one answer

13 Why uniqueness is not a technicality

Every firm's best decision depends on every other firm's, and every wage depends on every other wage. That is circular, so we must ask whether there is one self-consistent outcome or several.

> If the model had several equilibria, **no counterfactual would mean anything.** We measure "the effect of a tariff" by comparing before and after — but if the after has three possible outcomes, we cannot say which one the tariff produced, and the exercise collapses.

The rest of this part walks through the solution in the order the code does it. **Each step states its own reason for having exactly one answer.**

14 The order of solution, and the fact that makes it work

> **Market shares and markups depend only on costs, never on market size.** Look back at equations (1) and (3): E appears nowhere. It only scales levels afterwards. So once wages are fixed, every market can be solved *before* we know anybody's income.

That gives a clean, almost non-circular ordering:

Step	Solve for	How, and why it has one answer
1	shares, markups, prices	bracketed root find; *proved* unique
2	the input–output price loop	*proved* unique: a contraction
3	sector spending shares ε_{nk}	closed form; nothing to converge
4	expenditures E_{nk}, incomes X_n	two linear systems, solved exactly
5	wages w_n	the only genuine fixed point; verified

In words: Four of the five steps are either proved unique or have no freedom at all. Only the last one rests on numerical evidence.

15 Step 1: shares and markups in one market

Given wages, equation (8) gives every cost. Now solve one market.

15.1 Each group has exactly one best response

A parent with five affiliates seems to choose five numbers. **It is really choosing one.** Define the group's *output bundle* $X_g \equiv \sum_{i \in g} q_i^{\rho}$, and let B_g be the same sum over everybody else. Then $S_g = X_g/(X_g + B_g)$, so revenue depends on the group's five choices *only through the single number* X_g.

In words: Whether the group makes its bundle mostly in Colombia or mostly in Peru does not change its revenue at all. Only the total matters. We verify this in the code by literally reshuffling output across a group's plants at constant X_g: revenue agrees to 15 decimal places.

Now two facts about that single number:

- **Revenue rises but flattens** in X_g — each extra unit brings in less than the last, because you are already much of the market. (*Concave*: a hill that gets less steep as you climb.)
- **Cost rises and steepens.** Producing a bigger bundle as cheaply as possible means leaning on your less efficient plants, so the average worsens:

$$C_g(X) \;=\; X^{\frac{\sigma}{\sigma-1}}\,K_g^{-\frac{1}{\sigma-1}}, \qquad K_g \;\equiv\; \sum_{i\in g} c_i^{1-\sigma}. \tag{16}$$

The exponent exceeds 1, so cost is *convex*.

K_g deserves a name: the group's **cost-efficiency index**. It is the only thing about a group's internal cost structure that matters — two groups with the same K_g behave identically no matter how many plants they have.

Why this step has one answer. Profit is (rising, flattening revenue) minus (rising, steepening cost). Concave minus convex is **strictly concave**: a single smooth hill with exactly one summit. There is no second peak to get stuck on and no flat ridge. So each group has **one** best response — and the "set the derivative to zero" step of equation (12) is not merely necessary but *sufficient*.

We check this without using any formula: take two random production plans, average them, and confirm profit at the average always beats the average of the profits. Across 400 markets, every group, five random chords each: **zero violations**.

One loose end. Could a firm optimally produce nothing, putting a corner on the hill? No. As $q_i \to 0$ revenue falls like q_i^{ρ} while cost falls like q_i, and $\rho < 1$, so revenue falls more slowly. Producing a little always beats producing nothing. Every equilibrium is interior and the calculus always applies.

15.2 The market has exactly one consistent outcome

Each group only needs one summary of what everyone else does: the crowding measure A from (7). Given A, a group's share solves

$$\underbrace{S_g\,\mu(S_g)^{\sigma-1}}_{\equiv\;\psi(S_g)} \;=\; \frac{K_g}{A}. \tag{17}$$

Why this step has one answer. Because $\sigma > \eta$, the markup μ *rises* with share, so ψ is **strictly increasing** from 0 and never turns back. Therefore each group's share is uniquely determined by A, and is **strictly decreasing** in it — the more crowded the market, the smaller your slice.

Now impose $\sum_g S_g(A) = 1$. Every term falls as A rises, so the total falls. When A is tiny the total is near G, the number of groups, which exceeds 1; when A is huge the total is near 0. **A strictly falling curve that starts above 1 and ends below 1 crosses the level 1 exactly once.** That single crossing is the equilibrium.

Notice where $\sigma > \eta$ did the work: without it, ψ need not be increasing and the crossing need not be unique. This is why the code asserts it.

From that single A, everything else follows by substitution with no choices left:

$$A \longrightarrow S_g \longrightarrow \mu_g \longrightarrow p_i = \mu_g c_i \longrightarrow r_i \longrightarrow q_i.$$

So the market equilibrium is unique in *every* object. And because we construct it rather than merely proving it exists, existence comes free.

15.3 Why the code bisects

Bisection is not a stylistic choice — it is the algorithm the mathematics hands you. Both loops search along a strictly monotone function, so bisection cannot land on a "different" solution: there is no different solution.
The obvious alternative, repeatedly guessing shares and updating them, is *not* guaranteed to converge here. It hung during development. That is why the code does not use it.

16 Step 2: the input–output price loop

Now the loop the intermediates created. A firm's cost depends on the prices of the inputs it buys, and those prices are set by other firms whose costs depend on *their* inputs. Where does it settle?

Why this step has one answer. Exactly one place, and this one is a theorem. The argument is two lines once set up properly.

First, the market price index is **homogeneous of degree one in costs**. Multiply every cost in a market by λ: then every K_g is multiplied by $\lambda^{1-\sigma}$, so A is too, every share S_g is *unchanged*, and $P = A^{1/(1-\sigma)}$ is multiplied by exactly λ. In logs: a 1% rise in all input costs raises the price index by 1%.

Second, from equation (8), $\ln a$ depends on $\ln P^{\mathrm{IO}}$ with coefficient ν_k, and from (10) the weights $\omega_{k'k}$ sum to one.

Chain them. The map T taking the vector of log prices to itself has every row sum of its derivative bounded by $\max_k \nu_k < 1$. That is a **contraction** in the sup norm. By Banach's fixed point theorem it has **exactly one** fixed point, simple iteration converges to it *from any starting point*, and the error shrinks by a factor ν each round.

That last clause is a testable prediction, not just reassurance: reaching a tolerance ϵ should take about $\log \epsilon / \log \nu$ iterations. The code checks it:

ν	Iterations used	$\log(10^{-14})/\log \nu$
0.30	27	27
0.50	45	47
0.70	87	91

In words: The theory predicts the runtime almost exactly. That is about as direct a confirmation of the contraction argument as one could ask for.

Where it would break. Only if $\nu_k \geq 1$ — a sector spending its entire cost on intermediates and nothing on labour. That is economically impossible, and it is the same condition that keeps the input–output system productive in any Leontief-style model.

17 Step 3: how spending splits across sectors

Step 2 delivers each market's price index P_{nk}. Equation (5) then gives the sector shares ε_{nk} directly.

Why this step has one answer. It is an explicit formula — no search, no iteration, nothing to converge. And the shares add to one by construction, so no income leaks.

18 Step 4: expenditures and incomes

Now the circularity that looks worst, and which the intermediates made worse: incomes depend on profits, profits depend on market size, market size depends on incomes — *and* on how much firms buy from each other.

It dissolves, because with shares already known everything here is **linear**. Each firm's purchases of intermediates are a fixed fraction of its sales, so expenditure in a cell is final demand plus a linear function of expenditure everywhere else:

$$\mathbf{E} \;=\; \boldsymbol{\varepsilon} \circ \mathbf{X} \;+\; \mathbf{B}\,\mathbf{E} \qquad \Longrightarrow \qquad \mathbf{E} \;=\; (\mathbf{I} - \mathbf{B})^{-1}\,\boldsymbol{\varepsilon} \circ \mathbf{X}, \tag{18}$$

and then profits and tariff revenue are linear in $\mathbf{E}$, so equation (14) becomes one more linear system:

$$\big(\mathbf{I} - \mathbf{M}\big)\,\mathbf{X} \;=\; \mathbf{w} \circ \mathbf{L}. \tag{19}$$

Why this step has one answer. Two linear systems, solved *exactly* by elimination — not iterated to a tolerance. A linear system has one solution whenever its matrix is invertible, and here that has a plain economic meaning: the input–output loop must not feed more than 100% of an industry's sales back into itself as inputs (that is $\mathbf{B}$), and the profit-and-tariff loop must not feed back more than 100% of income (that is $\mathbf{M}$). Both hold whenever $\nu_k < 1$.

19 Step 5: wages — the only genuine fixed point

Everything so far was conditional on wages. Now compute labour demand from (13) and compare it with L_n. Where labour is scarce, raise the wage; where it is slack, lower it. Repeat.

Why this step has one answer. Here we are honest: this step is verified, not proved.

For the *market* equilibrium we have a genuine proof (Step 1). For wages we do not. The standard argument for uniqueness in trade models relies on *gross substitutes*, which can fail once markups vary with market share — a big firm's response to a wage change is not guaranteed to have the textbook sign.

So we test it hard. **110 random starting wage vectors** spread over a factor of about e^6, across five economies that differ in size (4 to 7 countries, 3 to 5 sectors), in the elasticity η (1 to 2.5), and in tariffs (0 to 40%).

Economy	Starts	Max wage gap	Worst labour residual
N=4, K=3, η=1, ν=0, free trade	30	2.7×10^{-13}	2.7×10^{-13}
N=4, K=3, η=1, ν=0.45, t=15%	30	1.7×10^{-12}	9.9×10^{-13}
N=5, K=4, η=1.8, ν=0.45, t=25%	20	3.9×10^{-13}	3.1×10^{-13}
N=6, K=5, η=2.5, ν=0.60, t=10%	15	9.0×10^{-13}	4.2×10^{-13}
N=7, K=3, η=1, ν=0.45, t=40%	15	3.7×10^{-13}	3.4×10^{-13}

Every start reaches the same wages, and the last column confirms every one of them genuinely clears the labour market — so a "match" cannot be two failures agreeing with each other. We report this as strong numerical evidence and we do not claim a theorem.

Two structural facts make this step well behaved, and both are checked:

- **Only relative wages matter.** Scaling every wage by any factor leaves excess demand unchanged (verified to 4×10^{-15}). So we normalise $w_1 = 1$ and lose nothing.
- **One condition is redundant** by Walras' law (15), so there are $N-1$ equations in $N-1$ unknowns — exactly as many as needed.

20 Everything we verified numerically

The code re-checks every claim above rather than asking you to trust it.

Check	Result
Accounting, at arbitrary non-equilibrium wages	
Walras' law, $\sum_n w_n Z_n = 0$	2.6×10^{-16}
Goods market clearing, $\sum_g S_g = 1$	6.7×10^{-16}
Revenue = labour + intermediates + tariffs + profits	3.2×10^{-16}
Each country's budget, equation (14)	2.4×10^{-16}
World budget: final spending = factors + tariffs + profits	2.2×10^{-16}
Intermediate demand = intermediate purchases	2.8×10^{-16}
Homogeneity: scaling all wages changes nothing	2.0×10^{-15}
Labour markets clear at the solution	2.0×10^{-13}
Uniqueness	
Revenue depends only on the bundle X_g	8.2×10^{-16}
Revenue concave, cost convex (2,000 draws)	0 violations
Group profit concave in own quantities (200 markets)	0 violations
ψ strictly increasing (500 draws)	0 violations
Market equilibrium, 200 starts across 5 structures	all identical, 8.9×10^{-10}
I–O price loop: iterations vs $\log \epsilon / \log \nu$	27/27, 45/47, 87/91
General equilibrium, 110 starts across 5 economies	all identical, $\leq 1.7 \times 10^{-12}$

Part IV
Taking the model to data

21 The denominator problem

This caused a genuine error in an earlier draft, so it is worth going slowly.

We wanted to check the model against Figure 6 of the stylised-facts document, which reports a concentration index of 0.192, rising to 0.215 when affiliates are grouped by parent. The model produces a concentration index too. So we compared them. **That was wrong: the two count different things.**

	What is in the denominator
Model's S_g	Everything sold in *one destination market* — including that country's own producers and exporters from everywhere else.
Figure 6's index	Only *Latin American export value in that product*, pooled across destinations — because customs data from nine LAC countries is all we observe.

Both work out to "about five effective firms", which is exactly why the comparison looked reasonable. But one counts all suppliers to Brazil and the other counts all LAC exporters of a product.

22 The fix, and what it revealed

We cannot rebuild the model's denominator in the data — we cannot observe Brazil's domestic producers. So we do the reverse: **simulate the customs records the model implies, and run Figure 6's own procedure on them.** Then both sides are the same object by construction. That procedure keeps only observed origins, pools across destinations within a product, aggregates to the chosen firm concept, rescales within the observed sample, and value-weights across products.

The rescaling step cancels the fringe out. Running the model with a larger and larger fringe of small competitors:

Size of fringe	λ (observed share)	Structural index	Measured index
0	1.000	0.244	0.079
16	0.548	0.092	0.080
256	0.133	0.010	0.107

The structural index collapses by a factor of 24; the measured one barely moves. **Figure 6 is essentially blind to the fringe, so it can never be used to measure it.** The original plan — calibrate the fringe to 0.192/0.215 — could not have worked however carefully it was executed.

The correct identification uses two moments for two objects: λ, the observed share of what the destination buys, pins down the **size of the fringe**; the measured index pins down **how many LAC exporters there are and how unequal they are**. They must be matched jointly.

Once the comparison is like-for-like, the model's grouping effect overshoots the data by about 1.6× rather than the 3.2× previously recorded. Most of the apparent mismatch was the denominator mistake, not the model.

Part V

What the model is for

23 Who owns the profits

Home country profit income from a market is $\sum_g \theta_{gn}\Pi_g$, which using equation (2) is

$$\Pi_H = \frac{E}{\sigma}\Big[\underbrace{\sum_g \theta_g S_g}_{\text{what any model gives}} + \left(\tfrac{\sigma}{\eta} - 1\right) \underbrace{\sum_g \theta_g S_g^2}_{\textbf{the paper}} \Big]. \tag{20}$$

The second sum is an **ownership-weighted Herfindahl**. It is neither the average ownership share nor the average concentration, and it needs firm-level ownership to compute.

24 Why the average ownership share is not enough

One example settles it. A market has four firms with shares $40, 30, 20, 10\%$. The home country owns half the market either way:

	Home owns	Home's share	$\sum_g \theta_g S_g^2$
Case A	the 40% and 10% firms	50%	$0.16 + 0.01 = 0.17$
Case B	the 30% and 20% firms	50%	$0.09 + 0.04 = 0.13$

In both cases the market's concentration is 0.30 and the ownership share is 50%.

Anyone with country-level data sees "50% ownership, concentration 0.30" in both cases and computes $0.5 \times 0.30 = 0.15$.
The truth is 0.17 in Case A and 0.13 in Case B — a **31% difference** that country-level data cannot see. The whole difference is *which* firms you own, not *how much* you own.

Formally, writing $\bar\theta$ for the ownership share and H for concentration,

$$\sum_g \theta_g S_g^2 = \underbrace{\bar\theta\, H}_{\text{what country data gives}} + \underbrace{\mathrm{Cov}(\theta, S)}_{\text{what we add}}. \tag{21}$$

25 The data ladder

Across many markets the same idea splits three ways, because ownership can also correlate with concentration *across* markets:

$$\underbrace{\bar\theta \cdot \bar H}_{\text{(1) naive}} + \underbrace{\mathrm{Cov}\big(\bar\theta_m, H_m\big)}_{\text{(2) between markets}} + \underbrace{\textstyle\sum_m w_m \,\mathrm{Cov}_m(\theta, S)}_{\text{(3) within markets}}.$$

What you have	What you can compute
Country ownership share + a published concentration index	(1) only
+ ownership share market by market	(1) + (2)
+ firm-level parent identity **(us)**	(1) + (2) + (3)

That table *is* the contribution, in one picture. Every competing paper sits on the first row.
Note carefully what is being claimed. This is a statement about **measuring who earns what** — an accounting identity, and a solid one. It is *not* yet a statement about optimal policy. Section 26 shows the policy implication is genuinely open, and may point the other way.

Two traps when reporting it. Both were found by testing, not by thinking. First, the ratio "truth ÷ naive" does **not** tend to 1 in a competitive market — it tends to the size premium of home-owned firms, because numerator and denominator vanish together. So it must always be reported next to the *level.* Second, a positive "within" term appears even with *no* ownership sorting at all, simply because multinationals are bigger than the fringe. That is the multinational-size channel, and reporting it as ownership sorting would overstate the contribution.

26 What the model says about tariffs

A tariff does four things to the country imposing it. It raises revenue. It raises consumer prices. It cuts the profits of firms *that country's own residents own.* And — only in general equilibrium — it lowers foreign wages, and so lowers the price the country pays for imports. That last one is the terms-of-trade gain, and it exists only because costs contain wages.

26.1 A result that runs against the project's hypothesis

The project's working hypothesis is Brecher–Bhagwati: if you own the firms you are taxing, taxing them takes from your own pocket, so your optimal tariff should be *lower.* We tested that in the general equilibrium model, letting country 1 own a rising fraction of the foreign multinationals it taxes.

Country 1 owns	dW/dt at free trade	Optimal tariff
0%	−0.209	below −0.90
25%	−0.146	below −0.90
50%	−0.012	0.00
75%	+0.190	+0.20
100%	+0.459	+0.55

The optimal tariff RISES with home ownership. The hypothesis has the sign backwards in this model.
The result is monotone across all five ownership levels, and it comes from two independent calculations: the marginal welfare effect at free trade (which needs no grid and cannot hit a corner) and a direct search for the optimal tariff.

26.2 Why — and it is not a bug

Brecher–Bhagwati is still there. It is simply outweighed by a second force that exists *only* because these are multinationals:

1. A tariff lowers demand for foreign goods, so it lowers foreign **wages**. (Verified: average foreign wage falls from 0.684 to 0.630 as the tariff goes from 0 to 40%.)

2. The firms country 1 owns **produce abroad**, hiring that foreign labour — look again at the cost function (8).
3. So their wage bill falls and their **profits rise**. Those profits flow to country 1.

> **Owning a foreign *exporter* is not the same as owning a foreign *affiliate*.**
> The classic result concerns ownership of firms whose costs you cannot influence. Here, a tariff is partly a device for depressing foreign wages — and if you own the capital that hires foreign labour, you capture those rents. You collect the terms-of-trade gain twice: once as a consumer paying less, and again as the shareholder of a firm whose costs just fell.
> This is a genuine economic mechanism, not an artefact, and as far as we can tell it is not in the ownership-and-tariffs literature, which works with country-level equity positions rather than with affiliates that hire foreign workers.

26.3 What this means for the project

Three things, and they should be stated plainly rather than smoothed over.

- **The income result is untouched.** Equation (20) and the data ladder are accounting identities about *who earns what*. They are correct and they remain the measurement contribution.
- **The policy claim is not established, and may have the wrong sign.** "Home ownership lowers the optimal tariff" does not follow from this model. It is one force among two, and in this calibration it loses.
- **Which force wins is an empirical question with a clear answer route.** It depends on α_k — how much of an affiliate's cost is headquarter labour rather than host labour — and on where the affiliates you own actually produce and sell. If US-owned Latin American affiliates ship mostly *back* to the US, the Brecher–Bhagwati force is strong; if they mostly serve third markets while hiring Latin American labour, the wage channel dominates.

> This is exactly why tabulating *where multinational affiliate exports actually go* is the highest-value empirical task on the project. It is cheap, and the **sign of the headline result depends on it.**

26.4 Two smaller cautions about magnitude

The tariff base is the *customs value*, not the shelf price, so revenue is deflated by the markup: taxing a high-markup supplier collects little per unit of damage. And only oligopoly lets a tariff extract profits at all — with constant markups a tariff is passed through completely and extracts nothing. Both effects are invisible in models without variable markups.

Part VI
Does the model explain the stylized facts?

27 The calibration

Before any result, here is every number that goes into the model and where it comes from. Three kinds, and the difference is what makes the exercise meaningful.

Parameter	Value	Kind	Where it comes from
σ (within sector)	5.0	external	trade elasticity 4–6
η (across sectors)	1.0	external	Cobb–Douglas outer nest
distance exponent	$1/(\sigma-1)$	external	makes gravity elasticity $= -1$
ν (intermediate share)	0.55	external	manufacturing I–O share
ω own-sector	0.45	external	I–O tables are diagonal-heavy
α_k (HQ share)	0.10–0.55	external	rises with complexity
$\gamma_{h\ell}$ (MP friction)	1.18	external	Ramondo–Rodríguez-Clare
MNE productivity edge	solved	**internal**	set to match **Fact 1**
HQ capability gradient	solved	**internal**	set to match **Fact 2**
country productivity	2.2 / 1.0	normalised	fixes relative wages
w_1	1.0	normalised	numeraire

Only two parameters are fitted, and both to stylized facts. Everything else is either taken from outside or is a pure normalisation. That is what makes Facts 4, 5 and 6 genuine out-of-sample tests: nothing in the calibration tells the model anything about concentration, about local firms' response to multinational entry, or about distance.

Health warning, and please act on it. The "external" values above are *standard magnitudes for this literature*, not transcriptions from specific tables in specific papers. Before anything goes into a paper, each one should be checked against its source and replaced with the published estimate. They are collected in a single `CALIB` block at the top of the code precisely so that this is a five-minute job rather than a treasure hunt.

27.1 One calibration choice worth understanding

The distance exponent is set to $1/(\sigma - 1)$. That is not a free choice dressed up: delivered cost enters demand with elasticity $(1 - \sigma)$, so this exponent makes the elasticity of *trade* with respect to distance exactly -1, which is the central estimate of the gravity literature.

It gives a free diagnostic. When the model's own gravity regression is run on model-generated data, ordinary firms should show a distance gradient near -1. They do: -0.93. The target is hit without being fitted to that regression.

This also reframes Fact 6. The model says -0.93; the gravity literature says about -1; the document's Table 2 says $\mathbf{-0.164}$. **It is the data coefficient that is the outlier**, an order of magnitude below any standard gravity estimate — which is exactly the specification concern already flagged for that table. So Fact 6 should be judged on its *interactions*, which is what the fact is actually about, not on the level.

28 How this was tested, and how to read it

The six stylized facts are the empirical motivation for the whole project, so the right question is not "is the model plausible" but "does it actually produce them". Everything below comes from *running* the model (`src/stylized_facts.jl`), not from arguing about it.

> Section 27 listed the parameters: **two are fitted, both to stylized facts**, and everything else is external or a normalisation.
> That makes Facts 4, 5 and 6 genuine **out-of-sample predictions**. Nothing in the calibration tells the model anything about market concentration, about local firms' response to multinational entry, or about distance.

Three verdicts are used, and the difference matters:

Generated	The model produces the fact from primitives. Real success.
Input / target	The fact is fed in and comes back out. No explanatory content, but the model is consistent with it and can use it.
Failed	The model predicts the opposite sign. Informative, and must be said out loud.

29 The scorecard

#	Fact	Verdict	What it needs
1	MNEs are a large share of exports	target	endogenous MP entry
2	Foreign MNEs in complex goods	target (matched)	endogenous MP entry
3	Parents from a few countries	input	endogenous MP entry
4	Grouping by parent raises concentration	**generated**	—
5	MNE presence raises non-MNE exports	needs spillover	test for selection
6	Distance matters less for MNEs	**generated**	—

In words: Two facts are real model output, two are assumptions the model is consistent with, and two need a further ingredient. Crucially, the two it generates are exactly the two the argument rests on: concentration is a property of the *parent*, and multinationals are not ordinary exporters.

30 Fact by fact

30.1 Fact 1 — multinationals are a large share of exports

Verdict: calibration target. Which firms are multinational is an input. We choose the single productivity-edge parameter so that the resulting export share matches the data, and it does: the model gives 0.44–0.67 across origins against 0.47–0.74 in the data.

One small bonus falls out unasked: the model reproduces *"overwhelmingly foreign-owned"* without being told to, because foreign parents are the productive ones and so win the share even though domestic multinationals face no multinational-production friction.

To make it generated. Add an entry margin: a fixed cost of going multinational, so that $\gamma_{h\ell}$ and country productivity decide who becomes a parent. That same extension also delivers Fact 3.

30.2 Fact 2 — foreign multinationals specialise in complex goods

Verdict: calibration target — the second fitted parameter. This one is genuinely instructive, so here is the experiment. For each candidate strength of an "HQ capability" channel, the productivity edge is *re-calibrated* so Fact 1 still holds — the level is pinned and only the gradient moves.

Capability slope	Foreign share, by sector				Gradient
0.0	0.62	0.47	0.54	0.58	−0.03
0.8	0.46	0.41	0.57	0.70	+0.60
1.6†	0.43	0.49	0.73	0.87	+1.04
Calibrated (0.588)	**0.51**	**0.43**	**0.57**	**0.67**	+**0.43**
Data (Fig. 2)	0.52	0.52	0.63	0.70	+**0.43**

† At this slope the level calibration hits its bound (the edge would have to go negative), so this row no longer holds Fact 1 fixed. The relevant comparison is the first two rows.

With no capability channel the gradient is **essentially zero** (−0.03): the foreign share wanders with the productivity draws and has no systematic relationship to α_k.

There is a reason, and it is in the cost function. Look back at equation (8): a foreign affiliate pays its *parent's* wage on the α_k share of cost, and parents sit in high-wage countries. So raising α_k makes foreign ownership *more expensive* — a force pushing directly *against* the fact, which roughly cancels the productivity edge.
The fact therefore tells us something the current model is missing: the headquarter input is not merely costly labour, it is a **capability local firms cannot buy at any price**, and it is worth more where the sector is more HQ-intensive.

Next step (Layer 2). Model the HQ input as a capability rather than as a wage bill. A capability slope near 0.6–0.8 brackets Figure 2's gradient of +0.43, so the parameter is identified off the data rather than assumed.

30.3 Fact 3 — parents come from a few countries

Verdict: input. We assign headquarter countries, so the concentration comes back out unchanged. *Next step:* the same endogenous-entry extension as Fact 1 — one addition buys both.

30.4 Fact 4 — grouping affiliates by parent raises concentration

Verdict: generated, and this is the model's central success. Computed exactly the way Figure 6 computes it (see Part V):

	Each affiliate	Parent × country	Global parent	Ratio
Model	0.069	0.078	0.121	1.75×
Data	0.192	0.209	0.215	1.12×

The *ordering* is a prediction, not an input, and it comes out right. The model overshoots the size of the jump and undershoots the level — there are too many firms per market in the synthetic economy, which is a calibration target, not a defect.

This is more than a measurement artefact, and that is the whole point. Because the markup depends on the **group's** share, grouping affiliates by parent raises measured concentration *and true market power at the same time.* The fact and the mechanism are the same object.

30.5 Fact 5 — more multinationals, more exports by *local* firms

This is the fact the input–output linkages were added for, so it gets the most space. The experiment adds ten multinational parents to one sector and measures what happens to non-multinational export value.

Step one: do the linkages fix it on their own? No.

Intermediate share ν	Same sector	Other sectors	All sectors
0.00	−40.4%	−6.0%	−15.3%
0.30	−38.9%	−6.1%	−15.1%
0.55	−36.2%	−5.8%	−14.2%
0.70	−32.6%	−5.2%	−13.2%
Data	**positive**		

Going from no linkages at all to a 70% intermediate share moves the number by about eight points — real, but nowhere near enough. Two forces beat it:

- Business stealing is **direct** and strong ($\sigma = 5$), while the input-price benefit reaches a local firm only through $\nu \times \omega \approx 0.2$ of its cost.
- In **general equilibrium the new plants bid up local wages**, raising local firms' costs. That is why even *other* sectors lose: the wage effect outweighs their cheaper inputs.

That second point is worth pausing on. A partial equilibrium model would have shown other sectors *gaining* from cheaper inputs and would have reported the linkage story as a success. The general equilibrium closure is what reveals that the wage channel dominates. This is a case where the extra machinery changed the answer, not just the decimals.

Step two: so what *would* it take? Add a knowledge spillover — local productivity scaled by $(1 + \#\text{MNE plants in own country-sector})^{\text{spill}}$ — and find the elasticity that flips the sign.

Spillover elasticity	Same sector	Other sectors	All sectors
0.00	−36.2%	−5.8%	−14.2%
0.10	−7.2%	−4.6%	−5.3%
0.20	**+21.3**%	−4.2%	+3.4%
0.30	+42.2%	−3.5%	+10.3%

The sign flips between 0.10 and 0.20. So Fact 5 *is* reproducible, but it needs input–output linkages *plus* a same-sector productivity spillover with elasticity around 0.15–0.20.

That turns "the model fails Fact 5" into a measurable question — and the answer is uncomfortable.
The spillover the model needs is **horizontal**: within the same sector. But Javorcik (2004), the standard reference, finds horizontal spillovers to be zero or *negative*, with positive effects only through **backward** linkages to upstream suppliers. The channel the model requires is precisely the one that literature says is absent.
Two readings, and the paper should commit to one:

1. the spillover is real and larger than usually estimated; or
2. Fact 5 is substantially **selection** — attractive markets draw in multinationals and local exporters alike — rather than an effect of multinationals *on* local firms.

Reading 2 is the more consistent with the evidence, and the empirical identification cannot currently rule it out: origin × destination × product fixed effects do not absorb time-varying market shocks, and no column carries destination × product × year.
It is also cheap to test: add destination × product × year fixed effects to Table 1. If the coefficient survives, reading 1; if it collapses, reading 2 and the model was right all along.

30.6 Fact 6 — distance is a weaker barrier for multinationals

Verdict: generated, at the affiliate level — which is the level the document's Table 2 actually uses. The regression is run on model-generated data with origin, destination and sector fixed effects, so market size cannot masquerade as a distance effect.

Distance gradient	Non-MNE	MNE	MNE present at destination
Model (affiliate level)	−0.93	−0.78	−0.73
Data	−0.16	−0.12	−0.05

Both interaction signs come out positive and in the data's order, and *nothing in the calibration targeted this.* It falls out of $d_{\ell n}$ running from the *production* country rather than the headquarters — the export-platform structure taken from Tintelnot (2017).

Two qualifications, both worth stating. The magnitudes are far too large: −0.93 against −0.164. Part of that is the small closed world used here, but note the data coefficient is itself an order of magnitude below any standard gravity estimate, which is the specification concern already flagged for Table 2. And at the *parent* level the "present at destination" term flips sign, because a parent with a plant at the destination already holds a large share there, so its markup is high and its distant plants are held back — cannibalisation pushing against the platform effect.

That gives Table 2 a sharp, falsifiable test. Re-run it with **firm fixed effects** and at the **parent level**. The model predicts the attenuation survives at the affiliate level and weakens at the parent level. If instead it vanishes with firm fixed effects, the fact was about firm size all along, not about networks.

31 Next steps, in the order that buys the most

1. **Fact 5 is the real anomaly.** Add input–output linkages so multinationals supply cheaper inputs to local exporters. Until then the model predicts business stealing and the data say the opposite. This is the only outright contradiction, so it goes first.

2. **Fact 2 needs the capability channel** (Layer 2): make the HQ input something local firms cannot buy, rather than merely expensive labour, and identify its strength off Figure 2.

3. **Facts 1 and 3 become generated** once multinational entry is endogenous — a fixed cost plus $\gamma_{h\ell}$ and country productivity deciding who becomes a parent and where. One extension buys both facts.

4. **Endogenous plant location** (Tintelnot's discrete choice) would turn Fact 6 from reproduced into explained. Hardest, and least urgent.

Part VII

Limits, and reference material

32 What this model does not do

- **Location choice is exogenous.** Which countries a multinational operates in is taken from the data, not chosen. See the caveat in Part IV.
- **No entry or exit.** The set of competitors in a market is given.
- **One factor.** No capital, land, or intermediate inputs.
- **General equilibrium uniqueness is verified, not proved.** The market-level result is a theorem; the wage-level result is numerical evidence from 60 random starts.
- **It does not reproduce Stylised Fact 5.** In the data, exports by *non*-multinational firms *rise* where multinationals are present. This model predicts the opposite. Something outside it — market growth, supply-chain linkages, or simply that attractive markets attract everyone — is doing that work. This is the one fact the model currently fails, and it should be stated openly.

33 Glossary of every symbol

Symbol	Meaning	Where it comes from
n, m	country	—
k	sector / product	—
a	affiliate (a plant)	ORBIS/D&B
g	group: the global ultimate parent	ORBIS/D&B
$h(a), \ell(a)$	HQ country and production country of affiliate a	data
L_n, w_n	labour endowment and wage	endowment / solved
X_n	country n's income = its spending	solved
σ	how easily buyers switch brands	calibrated
η	how easily buyers switch products ($\sigma > \eta \geq 1$)	calibrated
β_k	sector weight in demand	data
ρ	shorthand for $(\sigma - 1)/\sigma$	definition
E_{nk}	spending on sector k in country n	solved
α_k	headquarter input share	calibrated (Fact 2)
$\gamma_{h\ell}$	multinational production friction	calibrated
$d_{\ell n}$	iceberg trade cost	gravity data
$t_{\ell nk}$	tariff	policy
φ_a	affiliate productivity	calibrated
$a_{a,n}$, $c_{a,n}$	delivered cost before / after tariff	derived
s_i, S_g	variety and group market shares	solved
μ_g	markup, common across a group's affiliates	solved
Π_g	group profit	solved
θ_{gn}	fraction of group g owned by country n	ORBIS/D&B
K_g	cost-efficiency index $\sum_{i\in g} c_i^{1-\sigma}$	derived
X_g	group output bundle $\sum_{i\in g} q_i^{\rho}$	derived
A	market crowding; what the inner loop solves for	derived
P_{nk}, P_n	sector and aggregate price indices	derived
H	Herfindahl index $\sum_g S_g^2$	derived
λ	share of the destination's purchases we observe	customs / Comtrade

34 Where to find things

Everything is in one file: `mne_model.jl`. It is self-contained, needs no packages beyond the Julia standard library, and reproduces every number in this document. Share it as it stands.

Command	What it runs
`julia mne_model.jl`	everything, moderate replication
`julia mne_model.jl quick`	the headline results only
`julia mne_model.jl full`	heavy verification
`julia mne_model.jl core`	one market + the uniqueness proofs
`julia mne_model.jl ge`	general equilibrium + accounting audit
`julia mne_model.jl facts`	the six stylized facts

Its internal structure follows this document: Part 1 is one market, Part 2 the uniqueness proofs, Part 3 general equilibrium with input–output linkages, Part 4 the measurement operator, Part 5 the test economies, Part 6 the runner.

Other files	Contents
`scripts/verify/*.py`	independent Python versions, to catch coding errors
`docs/derivations.md`	all the algebra, terser than this document
`CLAUDE.md`	project status, open decisions, literature
`src/*.jl`	the earlier split-file version, kept for reference